

Probability – 2

Geometric probability and random points

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Theoretical minimum.

Def. A point is *uniform* in a region D of finite measure if $\Pr(X \in A) = \frac{|A|}{|D|}$ for every measurable $A \subseteq D$. Points are *exchangeable* if their joint distribution is invariant under permutations.

Theorem 3 (probability as a ratio, and conditioning). For uniform points the probability of an event is the ratio of measures in the product space, so a question about n random points in D is an integral over D^n . Conditioning on a part of the configuration often reduces the dimension: for points on a circle one conditions on the set of positions and uses exchangeability to compute the probability of a given cyclic order.

Theorem 4 (Crofton and Cauchy). The measure of lines meeting a convex body, normalised so that the measure of lines meeting a segment of length ℓ is 2ℓ , equals the perimeter. Hence the expected length of the projection of a convex body on a uniformly random direction is $\frac{\text{perimeter}}{\pi}$ in the plane, which is Cauchy's formula, and Buffon's needle problem is its immediate corollary.

Fact 2 (symmetry beats computation). Most olympiad problems about random points are solved without integrals: one finds a symmetry of the configuration under which the event maps to its complement, or counts cyclic orders, or applies linearity of expectation to indicators of “this point is a vertex of the hull”. Integrals appear only when the answer is not a rational number.

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1. Prove Theorem 4 in the plane, deduce Cauchy's formula for the perimeter and solve Buffon's needle problem; then compute the expected length of the projection of a segment onto a random line in $\mathbb{R}^n$.

2. (IMC 2020) Choose 2019 points independently and uniformly in the unit disc and let C be their convex hull. Which is more probable: that C is a triangle, or that C is a quadrilateral?
3. (IMC 2023) Let $n, k \geq 3$ and choose n blue and k red points uniformly and independently on a circle. Let F be the intersection of the convex hull of the red points with the convex hull of the blue points, and let m be the number of vertices of F , with $m = 0$ if F is empty. Find $\mathbb{E} m$.
4. (Mexmat, 2024) Five points are chosen uniformly and independently on a circle; let $ABCDE$ be the resulting pentagon, with vertices listed counterclockwise from the first chosen point, and let $K = AD \cap CE$. Find the probability that the area of ACB exceeds the area of ACK .
5. (Putnam 2024) Let Ω be the circle of radius 9 centred at the origin and let Δ be the disc of radius 1 centred at $(r, 0)$ with $0 \leq r \leq 8$. Two points P and Q are chosen independently and uniformly on Ω . For which r is the probability that the chord PQ meets Δ minimal?